

Rob Basnet

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EDUCATION

Gettysburg College

Bachelor of Science, Computer Science

Gettysburg, PA

May 2027

- **Coursework:** Data Structures & Algorithms, Distributed Systems, Operating Systems, Assembly, Machine Learning, OOP, Web Development

TECHNICAL SKILLS

Languages: Python, Java, JavaScript, TypeScript, C, C++, SQL

Frameworks: React, Spring Boot, FastAPI, Node.js, Express, GraphQL, Tailwind CSS

Tools: Docker, Git, Firebase, Vite, Axios, Framer Motion, Maven

ML / Data: scikit-learn, NumPy, pandas, TF-IDF, Random Forest, Isotonic Regression

PROJECTS

AI-Powered Resume Analyzer | *Python, FastAPI, scikit-learn, Docker, React* | [GitHub](#)

- Built a full-stack ATS resume analyzer parsing PDF, DOCX, and TXT files using pdfminer.six, python-docx, and chardet, enabling multi-format scoring from a single upload flow.
- Engineered an NLP scoring engine with TF-IDF vectorization and cosine similarity to extract JD keywords, surface skill gaps, and return structured match scores with actionable improvement suggestions.
- Designed the FastAPI REST backend with Pydantic validation, CORS configuration, and Uvicorn deployment, then containerized the full application with Docker for consistent cross-environment execution.

Soccer Match Prediction Platform | *Python, scikit-learn, React 19, Vite, Tailwind* | [GitHub](#)

- Trained 5 Random Forest models on Premier League data to predict match outcomes (W/D/L), expected goals, and betting markets (Over 2.5, Over 3.5, BTTS) using the football-data.org API.
- Applied Isotonic Regression probability calibration and odds-prior blending to generate full match payloads with win probabilities and top scoreline candidates, displayed through a React 19 dashboard built with Vite and Tailwind CSS.

SafeZone AI | *React, TypeScript, Node.js, Google Gemini API, Leaflet, Firebase* | [GitHub](#)

- Led frontend development for a 3-person emergency response web app, building the React and TypeScript interface with routing, real-time alert views, and Leaflet-based evacuation maps using React Router and Vite.
- Wired a Node.js/Express backend to Google Gemini (gemini-1.5-pro) to power an emergency guidance chatbot, handling async API calls, loading states, and error fallbacks across a unified multi-feature interface.

Premier League Club & Player Database | *Java, Spring Boot, React, Python, SCSS* | [GitHub](#)

- Built an end-to-end PL stats platform with a Spring Boot REST API, React frontend, and Python data pipeline serving a searchable database of 500+ players filterable by team, nation, and position.
- Wrote Python scraping and cleaning scripts to normalize player statistics from CSV sources into the backend, and built the SCSS frontend with multi-filter search in a 2-person Git workflow.

WORK EXPERIENCE

Gettysburg College

Resident Assistant Lead

Gettysburg, PA

Aug 2025 – Present

- Mentored 38+ residents through conflict resolution and community programming, organizing 5+ events per semester and increasing floor participation by 40%.

Gettysburg College

CALC AID Peer Learning Assistant

Gettysburg, PA

Sep 2024 – Present

- Tutored 10+ students weekly in calculus, breaking complex problems into structured frameworks that moved students from below-average to above-average performance.
- Taught reusable problem-solving methods for limits, derivatives, and integrals that improved student confidence and long-term retention across multiple exam cycles.

Association for Computing Machinery (ACM)

Member

Gettysburg, PA

Sep 2023 – Present

- Competed in hackathons building SafeZone AI using React, TypeScript, Node.js, and Google Gemini API in a 3-person team under time-constrained conditions.
- Participated in competitive coding challenges applying Python, Java, and C++ to sharpen algorithmic thinking and collaborative software design skills.